APPLIED ADVANCED OPTIMISATION

1. a) The problem can be formulated as

$$\begin{aligned}
\min_{p_{ik}} \quad & \sum_{k=1}^{D} C\left(\sum_{i=1}^{N} p_{ik}\right)\Delta t \\
s.t. \quad & \sum_{k=1}^{D} p_{ik}\Delta t \geq E_i, \quad i=1,\ldots,N, \\
& \sum_{i=1}^{N} p_{ik} \leq P_{\max}, \quad k=1,\ldots,D, \\
& p_{ik} \geq 0, \quad i=1,\ldots,N, \quad k=1,\ldots,D.
\end{aligned}$$

2 marks to the objective function, 2 marks for each constraint (three constraints). [8 marks]

b) The problem can be formulated as

$$\begin{aligned}
\min_{t_i} \quad & \sum_{i=1}^{4} a_i (T - t_i)^2 \\
s.t. \quad & t_i \geq t_i^{\min}, \quad i=1,\ldots,4, \\
& t_1 + t_2 + t_3 + t_4 = T, \\
& t_i - t_j \leq \delta, \quad i,j=1,\ldots,4,\ i \neq j.
\end{aligned}$$

2 marks for the objective function, 2 marks for each constraint (three constraints). [8 marks]

c) Comparing the graphs in the figure, we see that the linear placement (i) concentrates the free points in a small area, while the quadratic placement (ii) spreads the points over larger areas.

The linear placement penalizes all lengths with a constant slope generating both very short links, and very long ones.

The quadratic penalty function puts a higher penalty on long lengths relative to short lengths, while for small lengths the penalty is almost negligible. As a result, the maximum length is shorter, but we also have fewer short links.

The resulting distributions are typical of the ℓ_1 and ℓ_2 penalty functions.

[4 marks]

Answers

2. a) i) The feasible set is the interval $[1,2]$. The (unique) optimal point is $x^* = 1$, and the optimal value is $p^* = 1$. Figure 2.1 shows the objective function (solid/blue line), the constraint function (dashed/orange line), and the feasible set (shaded/grey area). The plot also indicates p^* (red/dot). [4 marks]

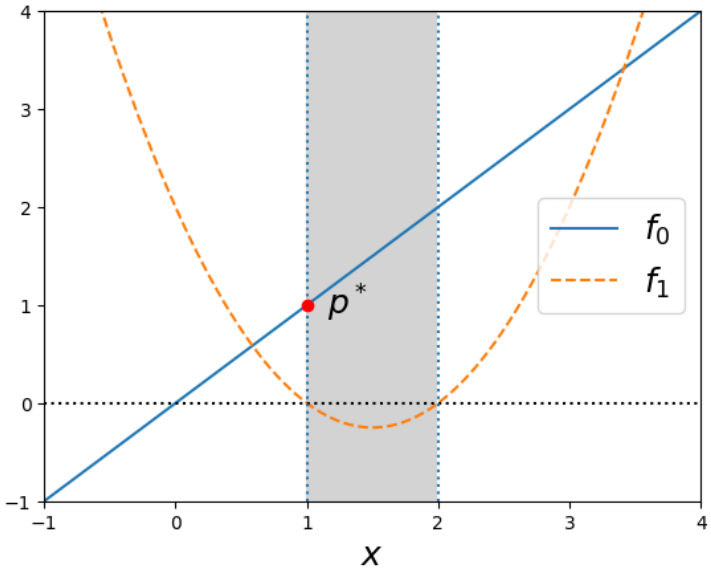


Figure 2.1 Plot for part a.i).

ii) The Lagrangian is

$$L(x,\lambda) = x + \lambda(x^2 - 3x + 2) = \lambda x^2 + (1 - 3\lambda)x + 2\lambda.$$

For $\lambda > 0$, the Lagrangian reaches its minimum at $\tilde{x} = \dfrac{3\lambda - 1}{2\lambda}$. For $\lambda \leq 0$, the Lagrangian is unbounded from below. Thus the dual function is

$$g(\lambda) = \begin{cases} \dfrac{-\lambda^2 + 6\lambda - 1}{4\lambda} & \lambda > 0, \\ -\infty & \lambda \leq 0. \end{cases}$$

[4 marks]

iii) The Lagrange dual problem is

$$\begin{array}{ll} \max\limits_{\lambda} & \dfrac{-\lambda^2 + 6\lambda - 1}{4\lambda} \\ s.t. & \lambda \geq 0. \end{array}$$

The dual optimum occurs at $\lambda^* = 1$, with $d^* = 1$. We can directly observe that strong duality holds (as it must because Slater's constraint qualification is satisfied). [4 marks]

iv) The perturbed problem is infeasible for $u < -\dfrac{1}{4}$, since $\inf\limits_x(x^2 - 3x + 2) = -\dfrac{1}{4}$. For $u \geq -\dfrac{1}{4}$, the feasible set is the interval

$$\left[\frac{3 - \sqrt{1 + 4u}}{2}, \frac{3 + \sqrt{1 + 4u}}{2}\right],$$

Answers

given by the two roots of $x^2 - 3x + 2 - u = 0$. For $u \geq -\frac{1}{4}$ the optimiser is $x^*(u) = \frac{3 - \sqrt{1+4u}}{2}$ (the closest point in the feasible set to the minimiser of f_0, which is unbounded below). In summary,

$$p^*(u) = \begin{cases} \infty & u < -\frac{1}{4}, \\ \frac{3 - \sqrt{1+4u}}{2} & u \geq -\frac{1}{4}. \end{cases}$$

Finally note that

$$\left.\frac{dp}{du}\right|_{u=0} = \left. -\frac{1}{\sqrt{1+4u}} \right|_{u=0} = -1,$$

which is equal to $-\lambda^* = -1$ found at point iii). [4 marks]

b) The Lagrangian is

$$L(x, \nu) = x^\top A^\top A x + (G^\top \nu - 2A^\top b)^\top x - \nu^\top h.$$

The KTT conditions are

$$2A^\top A x^* - 2A^\top b + G^\top \nu^* = 0, \qquad G x^* = h.$$

From the first equation we have

$$x^* = (A^\top A)^{-1} \left(A^\top b - \frac{1}{2} G^\top \nu^* \right).$$

Plugging this expression for x^* into the second equation gives

$$\nu^* = -2 \left(G (A^\top A)^{-1} G^\top \right)^{-1} \left(h - G (A^\top A)^{-1} A^\top b \right).$$

[4 marks]

Answers

3. a) i) Figure 3.1 shows the set

$$\{(x_1,x_2) \in \mathbb{R}^2 : (0,0),(1,0),(2,0),(0,1),(1,1),(2,2)\}.$$

indicated by 6 black dots, and its convex hull, indicated by the shaded region. [2 marks]

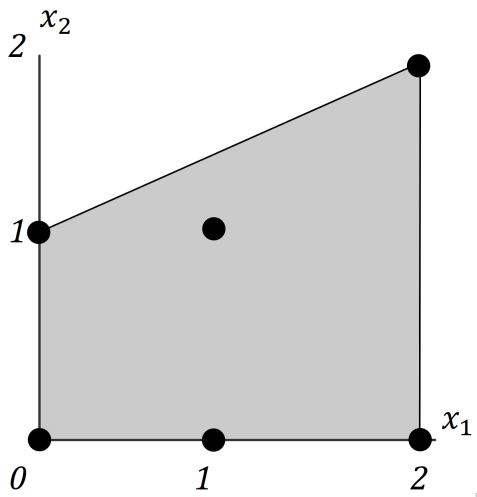


Figure 3.1 Plot for part 3.a.i).

ii) Figure 3.2 shows a set which has minimal points with respect to the cone $\mathbb{R}_-$. The minimal points are indicated in red. [2 marks]

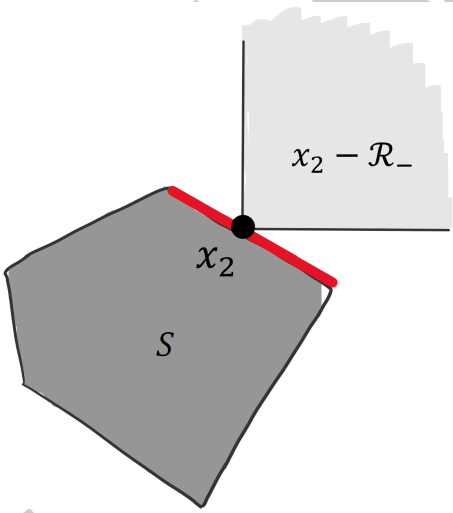


Figure 3.2 Plot for part 3.a.2).

b) i) We use the second-order condition.

$$f(x_1,x_2) = ax_1^2 + bx_1x_2 + cx_2^2 + dx_1 + ex_2 + f$$

$$\nabla f(x_1,x_2) = \begin{bmatrix} 2ax_1 + bx_2 + d \\ 2cx_2 + bx_1 + e \end{bmatrix},$$

$$\nabla^2 f(x_1,x_2) = \begin{bmatrix} 2a & b \\ b & 2c \end{bmatrix}.$$

Answers

The Hessian is positive semi-definite if the leading minors are nonnegative, thus if

$$2a \geq 0, \qquad 4ac - b^2 \geq 0,$$

then $f(x_1, x_2)$ is convex. [2 marks]

ii) We use the second-order condition.

$$f(x) = ae^{ax} + bx + c, \quad f'(x) = a^2 e^{ax} + b, \quad f''(x) = a^3 e^{ax}.$$

If $a \geq 0$, then $f''(x) \geq 0$ for all $x \in \mathbb{R}$, and f is convex. [2 marks]

iii) We use the second-order condition.

$$f(x) = \frac{ax+b}{cx+d}, \quad f'(x) = \frac{ad-bc}{(cx+d)^2}, \quad f''(x) = \frac{-2c(ad-bc)}{(cx+d)^3}.$$

If $2c(bc - ad) \geq 0$, then $f''(x) \geq 0$ for all $x \in D$, and f is convex. [2 marks]

iv) Consider two points (x_1, x_2) and (y_1, y_2) that belongs to S. We need to check if the point $(z_1, z_2) = (\theta x_1 + (1-\theta)y_1, \theta x_2 + (1-\theta)y_2)$, with $0 \leq \theta \leq 1$, belongs to S. Thus, we check

$$\begin{aligned} z_1^2 + z_2^2 &= \theta^2 x_1^2 + 2\theta(1-\theta)x_1 y_1 + (1-\theta)^2 y_1^2 \\ &\quad + \theta^2 x_2^2 + 2\theta(1-\theta)x_2 y_2 + (1-\theta)^2 y_2^2 \\ &= \theta^2(x_1^2 + x_2^2) + 2\theta(1-\theta)(x_1 y_1 + x_2 y_2) + (1-\theta)^2(y_1^2 + y_2^2) \\ &\leq \theta^2 + 2\theta(1-\theta) + (1-\theta)^2 = (\theta + 1 - \theta)^2 = 1, \end{aligned}$$

where for the inequality we have used $x_1^2 + x_2^2 \leq 1$, $y_1^2 + y_2^2 \leq 1$, and $x_1 y_1 + x_2 y_2 \leq \sqrt{x_1^2 + x_2^2}\sqrt{y_1^2 + y_2^2} \leq \sqrt{1}\sqrt{1} = 1$. [2 marks]

v) First note that $\mathbf{hypo}\, f = \{(x,t) : t \leq f(x)\} = \overline{\mathbb{R}^{n+1} \setminus \mathbf{epi}\, f} = \{(x,t) : f(x) \leq t\}$, thus the claim follows from proving that the epigraph is convex. To this end, let (x_1, t_1) be such that $f(x_1) \leq t_1$ and (x_2, t_2) be such that $f(x_2) \leq t_2$. Then

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta) f(x_2) \leq \theta t_1 + (1-\theta) t_2,$$

with $0 \leq \theta \leq 1$, where for the first inequality we have used the fact that f is convex. The above result shows that (x_3, t_3), with $x_3 = \theta x_1 + (1-\theta)x_2$ and $t_3 = \theta t_1 + (1-\theta)t_2$ is such that $f(x_3) \leq t_3$ for any $0 \leq \theta \leq 1$ and any (x_1, t_1) and (x_2, t_2). [2 marks]

c) The equivalent LP is

$$\begin{aligned} \min_{t,x} \quad & \mathbf{1}^\top t \\ \text{s.t.} \quad & -t \preccurlyeq x \preccurlyeq t, \\ & -\mathbf{1} \preccurlyeq Ax - b \preccurlyeq \mathbf{1}. \end{aligned}$$

Assume $x \in \mathbb{R}^n$ is fixed and optimise over $t \in \mathbb{R}^n$. The first constraint in the LP says that $-t_k \leq x_k \leq t_k$ for all k. Note that each of these constraints depends only on one t_k and that the objective function of the LP is the sum of the t_k. Hence, the objective function is separable and the optimum over t_k is achieved by $t_k = |x_k|$.

Clearly, if x is fixed, the optimal value over t of the LP is $||x||_1$. Thus, optimising over t and x simultaneously is equivalent to the original problem.

For the constraint $||Ax - b||_\infty \leq 1$, note that if $|a_k^\top x - b_k| \leq 1$ for all k, then $1 \geq \max_k |a_k^\top x - b_k| = ||Ax - b||_\infty$. [3 marks]

Answers

d) $z^*(u)$ satisfies the optimality condition

$$\frac{1}{u - f_0(z^*(u))} \nabla f_0(z^*(u)) + \sum_{i=1}^{m} \frac{1}{-f_i(z^*(u))} \nabla f_i(z^*(u)) + A^\top \nu = 0$$

for some ν. By direct comparison with the standard centrality condition we conclude that $z^*(u) = x^*(t)$ for

$$t = \frac{1}{u - f_0(z^*(u))}.$$

Conversely, for each $t > 0$, $x^*(t) = z^*(u)$ with

$$u = \frac{1}{t} + f_0(x^*(t)) > p^*.$$

[3 marks]

Answers